Next Phase: Generate Actual Experimental Data

Graph-Based Machine Learning for Attack-Path Vulnerability Prioritization

Objective

Develop an executable and independently reproducible experimental pipeline, replacing illustrative manuscript benchmarks with results calculated from actual model runs and attack-path simulations.

1. Graph construction

Define assets, vulnerabilities, identities, privileges, critical resources, and their relationships.

2. Dataset generation

Build a clearly labeled synthetic benchmark if authentic enterprise data is unavailable.

3. Model implementation

Train and evaluate GNN, XGBoost, and Random Forest using documented experimental procedures.

4. Attack-path simulation

Apply remediation budgets of 10, 20, 30, and 50 vulnerabilities.

5. Results validation

Calculate actual F1-scores, ROC-AUC, eliminated attack paths, residual exposure, and critical-asset reachability.

6. Repository publication

Upload the source code, experiment configurations, generated datasets, and resulting outputs.

Scientific integrity requirement

Report the values actually produced by the experiments, even when they differ from the illustrative figures in the manuscript. Clearly distinguish synthetic benchmark outcomes from empirical enterprise observations.

Browser-based workflow

GitHub files can be uploaded and maintained entirely through the website, without Git command-line operations. Generating experimental results still requires executing Python in an environment such as Google Colab or GitHub Actions.